

Problem Set 4

Problem 1. State the dimension theorem for a linear map $T: V \rightarrow W$.

Problem 2. Prove Rank-Nullity for your favorite $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$.

Problem 3. Use the spectral theorem on a symmetric 2×2 matrix.

Problem 4. Show $\dim \ker T + \dim \operatorname{im} T = \dim V$ without citing labels.

Problem 5. Let A be normal. Invoke the spectral theorem.

Problem 6. Explain why Theorem 3.22 and Rank-Nullity name the same fact.